Develop a CNN model for face recognition using the LFW dataset

To develop a Convolutional Neural Network for face recognition using the Labeled Faces in the Wild (LFW) dataset.

Algorithm
1. Import NumPy, TensorFlow and the LFW dataset loader.
2. Load face images and their corresponding identity labels.
3. Select individuals having a minimum number of face images.
4. Normalize the image pixel values.
5. Add a channel dimension to represent grayscale images.
6. Divide the dataset into training and testing sets.
7. Create a CNN model using convolutional layers.
8. Add max-pooling layers for feature reduction.
9. Flatten the extracted features.
10. Add Dense layers for learning facial features.
11. Add a softmax output layer representing different identities.
12. Train the model and evaluate its face classification accuracy.

Program

```
import numpy as np
import tensorflow as tf
from sklearn.datasets import fetch_lfw_people
from sklearn.model_selection import train_test_split
from tensorflow.keras import layers, models
tf.keras.utils.set_random_seed(15)
# Load LFW dataset
lfw = fetch_lfw_people(
min_faces_per_person=20,
resize=0.5,
color=False
)
# Prepare images
X = lfw.images.astype("float32") / 255.0
X = X[..., None]

y = lfw.target
num_classes = len(lfw.target_names)
# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
X, y,
test_size=0.25,
random_state=15,
stratify=y
)
# CNN model
model = models.Sequential([
layers.Input(shape=X.shape[1:]),
layers.Conv2D(32, 3, activation='relu', padding='same'),
layers.MaxPooling2D(),
layers.Conv2D(64, 3, activation='relu', padding='same'),
layers.MaxPooling2D(),
layers.Flatten(),
layers.Dense(128, activation='relu'),
layers.Dropout(0.4),
layers.Dense(num_classes, activation='softmax')
])
# Compile
model.compile(
optimizer='adam',
loss='sparse_categorical_crossentropy',
metrics=['accuracy']
)
# Train
model.fit(
X_train, y_train,
epochs=8,
batch_size=32,
validation_split=0.1
)
```

```
# Evaluate
loss, accuracy = model.evaluate(X_test, y_test, verbose=0)
# Predict

prediction = model.predict(X_test[:1], verbose=0).argmax(axis=1)[0]
print("Number of identities:", num_classes)
print("Test Accuracy:", accuracy * 100, "%")
print("Actual Identity:", lfw.target_names[y_test[0]])
print("Predicted Identity:", lfw.target_names[prediction])
```

```
Output of Program
Number of identities: Depends on dataset
Epoch 1/8
Epoch 2/8
...
Epoch 8/8
Test Accuracy: Training-dependent value
Actual Identity: ...
Predicted Identity: ...
```

Result

Thus, a CNN-based face recognition model was implemented using the LFW dataset. The model learned facial features and classified face images into their corresponding identity classes.